

Literature Review on Drift, Selection, and Recursive Retraining in LLM Text Ecosystems

Executive summary

The provided draft manuscript frames modern public text as an evolving “mixed human-AI” ecosystem: models generate drafts, humans and platforms select what gets published, and later models learn from that selectively visible record. In this view, two forces matter most: **drift** (finite-sample resampling that erodes low-frequency forms) and **selection** (publication, ranking, and verification that reweights the visible distribution toward what survives the filter), with later learners inheriting the resulting public conditional distribution via cross-entropy training. [filecite turn0file0](#)

Across the last decade of research, the strongest empirical and theoretical consensus is that **retraining generative models on their own outputs under “replace” dynamics** tends to reduce diversity and can induce “collapse” or distortion, often first visible in the tails of the distribution. ¹ At the same time, recent work shows that collapse is **not necessarily inevitable** in more realistic regimes that (a) retain or accumulate real data, (b) use verification/selection to curate synthetic data, or (c) incorporate correction mechanisms, but each mitigation typically trades off diversity, transparency, or representativeness against short-term performance and “viability.” ²

A key gap—directly aligned with the draft’s “artefact-learning vs process-learning” distinction—is that most collapse/mitigation studies optimize **output quality** or downstream accuracy, while fewer quantify how selective publication and hidden reasoning traces affect the learnability of *process* (search, verification, debugging, step-by-step reasoning). Process-supervision evidence strongly indicates that **training on intermediate steps can outperform outcome-only supervision** for complex reasoning, implying that ecosystems that systematically hide “failed attempts” may improve polished artefact imitation yet degrade process acquisition. ³

Methodologically, the literature is fragmented: (i) interpretable toy models (n-grams, Gaussian mixtures, linear models) isolate mechanisms but abstract away real web dynamics; (ii) deep-model iterative pretraining experiments demonstrate practical effects but are costly and confounded by data curation choices; (iii) platform-scale randomized experiments in computational social science demonstrate how ranking algorithms cause selection shifts in public discourse, underscoring that “selection” is not hypothetical. ⁴

Scope and framing for the user’s manuscript

The draft manuscript by Søren Riis ⁵ (affiliated with Queen Mary University of London ⁶) proposes a mechanism-level theory of “LLM text ecosystems” in which public text is recursively generated and re-ingested as training data. [filecite turn0file0](#)

Three manuscript claims are especially central for a literature review:

First, **neutral recursion produces drift**: finite-sample resampling causes rare words and higher-order phrases to be lost, with a Wright–Fisher-style limit at unigrams (the manuscript makes this analogy explicit). This aligns with empirical model-collapse evidence emphasizing early loss of distributional tails under iterative training on generated data. ⁷

Second, **trace filtering induces selection**: the visible corpus is shaped by curation/ranking/verification, producing a reweighted public conditional distribution (the manuscript formalizes this as acceptance-conditioned reweighting of next-token probabilities). This links directly to modern “verification” and selection pipelines for LLMs (self-consistency, verifiers, RLHF/RLAIF-like selection), which operationally reorder which trajectories survive into datasets and visible outputs. ⁸

Third, **inheritance is distributional, not architectural**: any sufficiently expressive next-token learner trained by cross-entropy on the same public environment tends to recover the environment’s conditional distribution, so downstream behavior is governed by the evolved public data distribution more than by model internals. This is consistent with a broader dataset-centric view of model quality: improvements (or degradations) can be driven by curation, deduplication, and filtering choices in web-scale corpora. ⁹

The manuscript’s primary method—an interpretable variable-order trigram model trained on public-domain fiction (e.g., Arthur Conan Doyle ¹⁰ via Project Gutenberg ¹¹)—is positioned as a “transparent toy ecosystem” where drift and selection effects are exactly computable and easy to simulate. The empirical illustration (Figure 3 in the draft, page 6) visualizes a recurring pattern seen across the broader literature: neutral recursion decreases higher-order support and diversity, while selection-like filtering can increase short-horizon viability/support at the cost of diversity.

Conceptual foundations across disciplines

A strength of the manuscript’s framing is that it connects recursive training to several mature theoretical traditions:

Population genetics provides a clean baseline for **finite-sample drift** in discrete generations (the Wright–Fisher model). In that setting, expected frequency can be a martingale while variance accumulates, making rare “alleles” (here: rare tokens/phrases) more likely to disappear under resampling. ¹² This analogy becomes especially plausible in unsmoothed or low-support language models where unseen events are assigned zero probability, directly exposing tail fragility.

Information theory offers a complementary lens via the **data-processing inequality**: post-processing cannot increase mutual information about an upstream source, and deterministic mappings cannot increase Shannon entropy for discrete variables. ¹³ The manuscript uses this to motivate why repeated “corpus → model → synthetic text → corpus” transformations risk losing information about the original data distribution, even if some resource-bounded “usable structure” can become more accessible through distillation.

Cultural evolution and cognitive science contribute the notion of **iterated learning chains**, where repeated transmission and learning amplify inductive biases and can drive convergence toward learnable/efficient structures. ¹⁴ This maps naturally onto recursive corpora: what survives publication and is easiest to learn may become overrepresented over time, even without any centralized intention. ¹⁵

NLP history grounds the manuscript's choice of n-gram models and back-off as transparent next-token predictors with well-studied sparse-data behavior: early work by Claude E. Shannon ¹⁶ established prediction-based measurements of English and inspired probabilistic language modeling; later work formalized back-off estimation for sparse counts (Katz; Kneser–Ney). ¹⁷

A compact way to relate these foundations to the manuscript's mechanism is:

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flowchart TB
    A[Public text environment] --> B[Sampling / scraping]
    B --> C[Fit next-token model]
    C --> D[Generate synthetic text]
    D --> E{Selection / filtering?}
    E -- No --> F[Neutral recursion: drift]
    E -- Yes --> G[Acceptance-conditioned publication: selection]
    F --> H[Next corpus]
    G --> H[Next corpus]
    H --> C

    subgraph Visibility
        I[Published artefacts] --> J[Future training data]
        K[Hidden traces / failed attempts] -. may be suppressed .-> J
    end
    end

```

This diagram matches the draft's core claim: what re-enters the environment can be unfiltered (drift) or filtered (selection), and the public record may exclude intermediate traces that are important for process learning. [filecite turnOf file0](#)

Annotated literature map of high-value sources

The works below are prioritized as (i) primary empirical studies of recursive retraining or synthetic-data contamination; (ii) theoretically grounded mechanism papers; (iii) dataset and ecosystem studies that operationalize “selection”; and (iv) foundational theory that tightens the manuscript's analogies. Unless otherwise noted, citations are in English and within the last decade; older works are included as foundational anchors.

Ilia Shumailov ¹⁸ et al. (2024). *AI models collapse when trained on recursively generated data*. *Nature* ¹⁹ **631: 755–759. ²⁰**

Summary: This paper formalizes and demonstrates “model collapse” under iterative training on model-generated data, emphasizing early loss of distributional tails and eventual convergence toward low-variance, “typical” modes. It argues collapse can occur even in nearly idealized settings (i.e., not merely due to function approximation error), and illustrates the phenomenon across multiple model families. ²¹

Methodology: Combined theoretical analysis with empirical demonstrations on Gaussian mixture models, variational autoencoders, and LLM-like settings. ²²

Key findings: Iterative self-consumption under “replace” dynamics erodes tails, producing irreversible defects and reduced diversity; the authors highlight this as a core risk for future web-scale training as synthetic text proliferates online. ²²

Limitations: The paper’s strongest results apply to the stylized “replace” regime; real-world training mixes data sources, and the mapping from toy theoretical models to frontier LLM pipelines depends on curation and accumulation assumptions. ²³

Relevance: Directly supports the manuscript’s drift-based “tail loss” concern and motivates studying drift and selection as separable forces in recursive ecosystems. [filecite turn0file0](#)

Sina Alemohammad ²⁴ et al. (2024). *Self-Consuming Generative Models Go MAD*. International Conference on Learning Representations ²⁵ (conference paper; arXiv:2307.01850). ²⁶

Summary: Introduces “Model Autophagy Disorder (MAD),” analyzing iterative self-consuming loops and showing progressive degradation in either precision (quality) or recall (diversity) when insufficient fresh real data is included each generation. The work stresses that selection toward high-quality samples can further trade off against diversity. ²⁷

Methodology: Analytical + empirical experiments primarily on state-of-the-art generative image models, comparing multiple loop designs with varying access to real data and varying bias toward quality vs diversity. ²⁷

Key findings: Without enough real data per generation, quality and/or diversity deteriorate inexorably over iterations, even when synthetic samples are filtered for quality. ²⁷

Limitations: Focuses mainly on images and particular loop families; transfer to text and to web-scale, multi-source corpora requires care. ²⁸

Relevance: Empirically complements the manuscript’s drift-selection trade-off: “quality filters” can improve apparent viability while accelerating diversity loss, mirroring Figure 3’s support-diversity tension. [filecite turn0file0](#)

Quentin Bertrand ²⁹ et al. (2024). *On the Stability of Iterative Retraining of Generative Models on their own Data*. ICLR 2024 (Spotlight; arXiv:2310.00429). ³⁰

Summary: Develops a theoretical framework for iterative retraining on mixed “clean + synthetic” datasets and identifies conditions under which training remains stable. It also empirically validates these conditions by iteratively training modern generative models over multiple steps. ³⁰

Methodology: Theoretical analysis plus iterative retraining experiments on normalizing flows and diffusion models (e.g., CIFAR-10 and FFHQ). ³⁰

Key findings: Stability depends on (a) the initial model being close enough to the true data distribution and (b) a sufficiently large proportion of clean (real) data in the training mix, echoing “mutation/immigration” in drift models. ³⁰

Limitations: Empirical validation is image-centric and controlled; it does not directly measure language diversity or public-record selection mechanisms. ³¹

Relevance: Provides formal support for a parameterized “clean fraction” in the manuscript’s recursion (analogous to keeping part of the corpus vs regenerating part), helping justify sensitivity analyses over mixture rates. [filecite turn0file0](#)

Matthias Gerstgrasser ³² et al. (2024). *Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data*. arXiv:2404.01413. ³³

Summary: Argues that “model collapse” conclusions are regime-dependent: replacing real data with

synthetic data tends toward collapse, but accumulating synthetic data alongside retained real data can avoid collapse. The work demonstrates this empirically for language models and other modalities, and provides an analytical argument showing bounded error under accumulation. ³⁴

Methodology: Empirical iterative pretraining of sequences of language models on text corpora under replace vs accumulate regimes, plus supporting experiments on other modalities (diffusion models, VAEs) and an analytically tractable linear-model framework. ³⁴

Key findings: Accumulation (archive grows) can prevent collapse observed under replacement; the difference is theoretically explainable via bounded test error under accumulation. ³⁴

Limitations: “Accumulation prevents collapse” does not guarantee preservation of long-tail linguistic richness or equitable representation; accumulation can still dilute tails in practice if synthetic growth dominates or if selection biases shape what accumulates. ³⁵

Relevance: Sharpens the manuscript’s message that “no single verdict” exists: dynamics depend on drift/selection and on whether data is replaced or accumulated, and this suggests explicitly modeling “archive growth” variants of the recursion. [filecite turn0file0](#)

Yanzhu Guo ³⁶ et al. (2024). *The Curious Decline of Linguistic Diversity: Training Language Models on Synthetic Text*. Findings of Association for Computational Linguistics ³⁷ : NAACL 2024. ³⁸

Summary: Shifts focus from accuracy to linguistic richness, proposing metrics for lexical, syntactic, and semantic diversity in recursive synthetic-text finetuning regimes. Finds consistent diversity decline across successive iterations, especially on creative generation tasks. ³⁹

Methodology: Recursive finetuning experiments on English NLG tasks; evaluation with newly developed multi-level diversity metrics. ⁴⁰

Key findings: Diversity decreases over iterations even when standard task performance is not the primary target; creative tasks are especially sensitive, suggesting tail contraction is a salient failure mode in practice. ⁴⁰

Limitations: Finetuning experiments may not fully represent pretraining-scale dynamics; diversity metrics can be sensitive to decoding strategies and prompt distributions. ⁴¹

Relevance: Direct empirical complement to the manuscript’s drift claim: tail erosion is not just theoretical, and it can be quantified beyond perplexity using diversity-focused metrics. [filecite turn0file0](#)

Matyas Bohacek ⁴² & Hany Farid ⁴³ (2023). *Nepotistically Trained Generative-AI Models Collapse*. arXiv:2311.12202 (later notes publication in an ICLR workshop). ⁴⁴

Summary: Demonstrates that retraining image generators on even small amounts of their own outputs can yield strong distortions that generalize beyond the prompts used for retraining, and that the models may not fully recover even after retraining on only real images. ⁴⁴

Methodology: Empirical study of iterative retraining (“nepotistic training”) in generative image synthesis, tracking qualitative distortions and recovery behavior under reintroduction of real data. ⁴⁵

Key findings: Self-generated contamination can cause rapid, hard-to-heal degradation; effects can generalize beyond the narrow training prompts, suggesting systemic distributional distortion rather than simple overfitting. ⁴⁶

Limitations: Image domain; does not disentangle drift vs explicit selection mechanisms, nor does it address “process trace” visibility. ⁴⁷

Relevance: Strengthens the manuscript’s cross-architecture, cross-domain argument that recursive self-training can create distributional pathologies that are not specific to transformers. [filecite turn0file0](#)

Nate Gillman ⁴⁸ et al. (2024). *Self-Correcting Self-Consuming Loops for Generative Model Training*. International Conference on Machine Learning ⁴⁹ (PMLR 235:15646–15677; arXiv:2402.07087). ⁵⁰

Summary: Proposes “self-correction” mechanisms to stabilize self-consuming loops: an idealized corrector that reweights samples toward the true data distribution, and practical approximations using expert knowledge (e.g., simulators, constraints). Demonstrates empirically that correction can prevent collapse even under extremely high synthetic-to-real ratios in a challenging generation task. ⁵⁰

Methodology: Theoretical analysis of loop stability with correction functions, plus experiments on human motion synthesis with high synthetic ratios. ⁵¹

Key findings: With correction, self-consuming loops can be made substantially more stable (the paper claims exponential stability improvements under assumptions), and empirical results show collapse avoidance at ratios as high as 100% synthetic relative to real. ⁵¹

Limitations: The feasibility of “expert-knowledge correctors” depends on domains with strong external constraints; transfer to open-ended text (where correctness is less checkable) is nontrivial. ⁵²

Relevance: Offers a concrete instantiation of the manuscript’s “selection/verification” story: selection need not be binary accept/reject, and the *shape* of the filter influences long-run stability and diversity. `filecite\turn0file0`

Yunzhen Feng ⁵³ et al. (2025). *Beyond Model Collapse: Scaling Up with Synthesized Data Requires Verification*. ICLR 2025 (Poster; arXiv:2406.07515). ⁵⁴

Summary: Argues that verification is a practical route to scaling synthetic-data training without collapse because it is often easier to judge good vs bad examples than to generate high-quality samples. Provides theory for verifier effectiveness in simple distributions and demonstrates empirically that verifiers can prevent collapse on tasks where naive synthetic training collapses. ⁵⁴

Methodology: Theoretical characterization using Gaussian mixtures and linear verifiers/classifiers, plus experiments including transformer-based eigenvalue computation and LLM-based news summarization. ⁵⁴

Key findings: Verifier-based selection can avert collapse even with imperfect verifiers; the authors propose proxy measures that correlate with downstream performance. ⁵⁴

Limitations: Verification assumes an available correctness signal (ground truth, constraints, or reliable evaluators); in open-ended prose, “verification” may collapse into subjective preference or popularity signals. ⁵⁵

Relevance: Closely parallels the manuscript’s acceptance-conditioned reweighting of next-token distributions and strengthens the claim that selection can stabilize “viability” while potentially reducing diversity and hiding underlying search. `filecite\turn0file0`

Guilherme Penedo ⁵⁶ et al. (2024). *The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale*. NeurIPS 2024 Datasets & Benchmarks (arXiv:2406.17557). ⁵⁷

Summary: Introduces FineWeb (15T tokens from 96 Common Crawl snapshots) and FineWeb-Edu (1.3T educational tokens), documenting and ablating key curation steps (deduplication, filtering, mixing). Shows that dataset design choices can materially improve downstream LLM performance, especially for knowledge/reasoning benchmarks in the educational subset. ⁵⁷

Methodology: Web-scale dataset construction + controlled ablations where models are pretrained on variants, isolating effects of filtering and dedup strategies. ⁵⁸

Key findings: Curation pipelines are not incidental; careful filtering/dedup yields measurably better pretrained models, and educational filtering can disproportionately boost knowledge- and reasoning-heavy evaluations. ⁵⁸

Limitations: The work optimizes for benchmark performance rather than explicitly tracking long-tail linguistic diversity or the visibility of intermediate traces; “better benchmarks” may not equal “better ecosystem health.” ⁵⁹

Relevance: Operationalizes “selection” at dataset scale—exactly the kind of curation pressure the manuscript theorizes—and provides an empirical vocabulary for describing filtering regimes (aggressive vs permissive, dedup depth, etc.). [filecite\turn0file0](#)

DataComp-LM (Jiacheng Li ⁶⁰ et al., representing the “DataComp-LM Consortium”) (2024). *DataComp-LM: In search of the next generation of training sets for language models*. NeurIPS 2024 Datasets & Benchmarks. ⁶¹

Summary: Proposes a controlled testbed for dataset experiments, offering standardized corpora from Common Crawl, training recipes, and broad downstream evaluations to compare curation strategies. Reports that model-based filtering is a key ingredient for assembling high-quality training sets under fixed compute budgets. ⁶²

Methodology: Benchmark + infrastructure paper with extensive baseline experiments at several model scales; controlled comparisons of dedup/filter/data-mix strategies. ⁶³

Key findings: Dataset design and filtering can yield large quality gains at lower compute; the published baseline demonstrates strong MMLU performance for open-data training regimes. ⁶¹

Limitations: Focuses on performance/compute; less emphasis on representational harms, trace visibility, or long-run recursive dynamics (it is a snapshot benchmark rather than an iterative retraining study). ⁶⁴

Relevance: Provides a concrete experimental platform to test the manuscript’s claims under controlled “selection” (filtering) regimes and to measure distributional drift/selection effects alongside performance. [filecite\turn0file0](#)

Jesse Dodge ⁶⁵ et al. (2021). *Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus*. EMNLP 2021. ⁶⁶

Summary: Documents the provenance and properties of the C4 corpus, showing that web-scale filters and blocklists can have unintended distributional consequences. Reports evidence of machine-generated text within C4 and shows that certain filtering steps disproportionately remove text from and about minority individuals (including dialect-associated English). ⁶⁶

Methodology: Dataset audit: domain/source breakdowns, content analyses, and assessments of the impact of filtering choices by examining removed versus retained text. ⁶⁷

Key findings: Filtering is itself a selection mechanism that reshapes the training distribution; it can remove undesirable content but also remove minority language and topic discussion, which can have downstream representational impacts. ⁶⁸

Limitations: Focuses on one major corpus snapshot and on audit methodology; it does not directly model recursive retraining dynamics over time. ⁶⁹

Relevance: Strongly supports the manuscript’s warning that “selection” is not benign: curation can clean artefacts while erasing minority discourse and rare forms, interacting with drift to accelerate tail loss. [filecite\turn0file0](#)

Emily M. Bender ⁷⁰ et al. (2021). *On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?* FAccT 2021. ⁷¹

Summary: Provides a broad critical framework for risks of large language models, emphasizing harms related to training data (bias, uncured web text), environmental costs, and the limits of scale as a research strategy. While not an iterative-training paper, it motivates why ecosystem-level dynamics and data governance matter for downstream social impacts. ⁷¹

Methodology: Conceptual and critical analysis synthesizing evidence from NLP practice, dataset construction, and social/ethical considerations. ⁷²

Key findings: LLM behavior is inseparable from data sources and filtering choices; opaque data pipelines

and unexamined scaling can amplify biases and externalities. ⁷²

Limitations: Not a primary empirical analysis of recursive retraining; its claims are broad and not tailored to specific drift/selection mechanisms. ⁷³

Relevance: Supplies normative and governance context for the manuscript’s descriptive theorems: even if drift/selection are “distributional,” their consequences include representational and societal harms when tails correspond to marginalized language or topics. `filecite{turn0file0}`

Mihai Nadas ⁷⁴ et al. (2025). *Synthetic Data Generation Using Large Language Models: Advances in Text and Code*. arXiv:2503.14023 (survey). ⁷⁵

Summary: Reviews techniques for generating synthetic text and code with LLMs, including prompt-based generation, retrieval-augmented pipelines, iterative self-refinement, and filtering/weighting strategies. Discusses benefits (cost, controllability) and failure modes (hallucinations, bias amplification, insufficient realism), and highlights verification in code as a strong mitigation lever. ⁷⁵

Methodology: Survey/review of methods spanning NLP and code generation, distinguished by generation pipelines and validation/verification strategies. ⁷⁶

Key findings: Verification and filtering repeatedly emerge as practical tools to improve synthesized datasets, but can distort distributions if used too aggressively or if verifiers encode narrow norms. ⁷⁷

Limitations: As a survey, it does not provide a single unified mechanistic theory of long-run ecosystem dynamics. ⁷⁸

Relevance: Helpful for positioning the manuscript’s theory against applied synthetic-data pipelines, and for identifying where “selection” enters in practice (filtering, weighting, execution-based checks). `filecite{turn0file0}`

Jason Wei ⁷⁹ et al. (2022). *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. NeurIPS 2022. ⁸⁰

Summary: Shows that prompting models to produce intermediate reasoning steps (“chain-of-thought”) can dramatically improve performance on multi-step reasoning tasks, especially at large scales. Establishes the modern importance of *visible traces* as both a capability and an evaluation object. ⁸⁰

Methodology: Prompting-based experiments on multiple reasoning benchmarks; comparison of direct-answer prompting vs chain-of-thought prompting and scaling effects. ⁸⁰

Key findings: Intermediate steps can unlock latent reasoning performance, and decoding strategies that encourage structured traces can change apparent competence. ⁸⁰

Limitations: Concerns inference-time prompting rather than training-time recursive ecosystems; does not analyze long-run distributional drift or selection in data. ⁸¹

Relevance: Supports the manuscript’s process-learning emphasis: if public ecosystems suppress traces, future models may lose access to an important substrate for learning and evaluation. `filecite{turn0file0}`

Xuezhi Wang ⁸² et al. (2023). *Self-Consistency Improves Chain of Thought Reasoning in Language Models*. ICLR 2023 (arXiv:2203.11171). ⁸³

Summary: Proposes sampling multiple reasoning paths and selecting the most consistent answer, improving performance relative to greedy chain-of-thought decoding. This is a concrete example of “selection” over multiple internal traces before producing a final output. ⁸³

Methodology: Inference-time sampling + aggregation; extensive benchmark evaluation on arithmetic and commonsense reasoning datasets. ⁸³

Key findings: Selection over diverse traces improves accuracy substantially on multiple benchmarks, suggesting that “filtering among candidates” is a powerful operator even without changing the base model.

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Limitations: The internal rejected traces are generally not published; thus, ecosystem-level implications depend on whether sampled traces are retained as training data or discarded. ⁸⁴

Relevance: Provides a practical, widely used analogue to the manuscript's lookahead/acceptance reweighting: ecosystems may preferentially publish only the selected branch, shaping future training distributions. `filecite turn0file0`

Hunter Lightman ⁸⁵ et al. (2023). *Let's Verify Step by Step*. arXiv:2305.20050. ³

Summary: Demonstrates that process supervision (step-level feedback) can outperform outcome-only supervision for training models to solve difficult math problems, and releases a large step-level feedback dataset (PRM800K). Presents verification as a scalable mechanism to improve reasoning reliability. ⁸⁶

Methodology: Human labeling of intermediate steps, training reward/verifier models, and evaluating on MATH; includes active learning to improve labeling efficiency. ⁸⁷

Key findings: Process supervision significantly improves training outcomes for multi-step reasoning; verifiers trained on step labels can reach strong performance on subsets of the MATH test set. ⁸⁶

Limitations: The domain is math reasoning with explicit step structure; generalization to open-ended tasks is unclear, and step traces may contain sensitive information that is often withheld from public corpora. ⁸⁸

Relevance: Direct empirical evidence for the manuscript's claim that filtering can harm process-learning: if ecosystems publish only final answers (artefacts) rather than full traces, the information needed for process supervision may disappear from the public record. `filecite turn0file0`

Long Ouyang ⁸⁹ et al. (2022). *Training language models to follow instructions with human feedback*. arXiv:2203.02155 (InstructGPT). ⁹⁰

Summary: Shows that fine-tuning with human feedback (supervised demonstrations + preference rankings + RLHF) can produce models that humans prefer over much larger base models. This provides a canonical "selection" pipeline in which outputs are ranked and filtered, shaping what behaviors are reinforced. ⁹⁰

Methodology: Instruction data collection, supervised fine-tuning, reward modeling from human rankings, RL optimization (RLHF), and human evaluation. ⁹⁰

Key findings: Preference-based selection can dramatically shift model behavior toward helpfulness/truthfulness as measured by human preference, independent of parameter count. ⁹⁰

Limitations: The approach depends on curated feedback distributions and the quality of reward modeling; many rejected outputs and intermediate attempts remain unpublished, potentially reducing process transparency. ⁹¹

Relevance: The manuscript's "selective publication" becomes concrete here: RLHF-like selection changes the visible distribution, and later learners trained on that visible distribution inherit its biases and omissions. `filecite turn0file0`

Ferenc Huszár ⁹² et al. (2022). *Algorithmic amplification of politics on Twitter*. PNAS 119(1): e2025334119 (arXiv:2110.11010). ⁹³

Summary: Uses large-scale randomized experimentation on a major social platform to quantify how algorithmic ranking changes what political content users see. Finds systematic amplification differences across political groups in multiple countries, illustrating that "selection" in public discourse can be measured causally. ⁹⁴

Methodology: Platform-scale randomized control group (~2M daily active accounts) exposed to reverse-chronological feeds, enabling causal estimation of algorithmic amplification; analyses of legislators' tweets across countries and U.S. news sources. ⁹⁴

Key findings: Personalized ranking can systematically amplify some political content relative to others, implying that "public text" is not a neutral sample but a selected, algorithmically mediated record. ⁹⁴

Limitations: Focuses on one platform and political content; does not analyze recursive model-training loops directly. ⁹⁵

Relevance: Strengthens the manuscript's sociotechnical motivation: selection pressures in public text are real and measurable, and they can bias the distributions that later models train on. ⁹⁶

Erwan Dujeancourt ⁹⁶ & Marcel Garz ⁹⁷ (2023). *The effects of algorithmic content selection on user engagement with news on Twitter*. The Information Society. ⁹⁸

Summary: Studies how switching from reverse-chronological to algorithmic timelines changes user engagement with news, emphasizing downstream behavioral effects of selection mechanisms. Provides evidence that ranking algorithms reconfigure attention and engagement, which can indirectly influence what content gets produced and retained. ⁹⁹

Methodology: Empirical analysis around a platform product change (timeline algorithm introduction), relating content selection to engagement outcomes (reported as a March 2016 switch in the abstract). ⁹⁹

Key findings: Algorithmic selection changes consumption and engagement with news, implying feedback onto production incentives and available "public text." ⁹⁹

Limitations: Downstream text consequences are indirect; study focuses on engagement rather than linguistic distributions or training-data implications. ¹⁰⁰

Relevance: Helps justify the manuscript's modeling choice to treat ranking/curation as a selection operator that shapes what enters the durable public record. ¹⁰⁰

Thomas M. Cover ¹⁰¹ & Joy A. Thomas ¹⁰² (2006). *Elements of Information Theory* (2nd ed.). Wiley. ¹⁰³

Summary: Standard reference for entropy, mutual information, and inequalities such as data-processing. Provides formal tools for reasoning about information loss under transformations—precisely the intuition used in the manuscript's "recursive loop as processing" discussion. ¹⁰³

Methodology: Theoretical textbook. ¹⁰⁴

Key findings: For Markov chains $X \rightarrow Y \rightarrow Z$, the data-processing inequality bounds $I(X; Z)$ by $I(X; Y)$, formalizing limits on recoverable information after processing. ¹⁰⁵

Limitations: Shannon information is not a complete account of "usable" information for bounded agents; bridging to computational constraints requires extra modeling assumptions. ¹⁰⁶

Relevance: Offers a rigorous foundation for the manuscript's information-theoretic motivation, and for future extensions that quantify information loss/gain under different publication filters. ¹⁰⁶

Claude E. Shannon (1951). *Prediction and entropy of printed English*. Bell System Technical Journal 30(1): 50–64. ¹⁰⁷

Summary: Classic early work on probabilistic prediction of English and entropy estimation, historically linking language modeling to next-symbol prediction. Serves as a foundational reference for treating language as a stochastic process with measurable predictability and redundancy. ¹⁰⁸

Methodology: Experimental + theoretical: human prediction experiments and derivations for entropy estimates. ¹⁰⁷

Key findings: Demonstrates that predictability improves with context length; motivates n-gram/Markov perspectives that the manuscript uses as a transparent analogue for next-token systems. ¹⁰⁸

Limitations: Letter-level modeling and human prediction are historically important but not directly comparable to modern tokenization and large-scale neural training. ¹⁰⁹

Relevance: Strengthens the manuscript's rhetorical lineage: studying next-token distributions with short-context models is not a regression but a return to interpretable roots. ¹⁰⁹

Simon Kirby, Tom Griffiths, & Kenny Smith (2014). *Iterated learning and the evolution of language*. *Current Opinion in Neurobiology* 28:108–114. ¹⁴

Summary: Reviews iterated learning as a mechanism of cultural evolution, covering mathematical models, agent-based simulations, and lab experiments. Emphasizes convergence toward learnable structures under repeated transmission, which parallels how recursive training can reshape distributions over time. ¹⁴

Methodology: Review article synthesizing computational models, theory, and experiments. ¹⁴

Key findings: Iterated learning can shift population-level behavior toward the prior/inductive biases of learners, even absent explicit communicative optimization. ¹¹⁰

Limitations: Primarily about human/agent language evolution; mapping to LLM ecosystems requires specifying how “learning biases” correspond to training objectives and data pipelines. ¹¹¹

Relevance: Provides a cross-disciplinary backbone for the manuscript’s “recursive ecosystem” framing and suggests additional modeling tools (iterated learning Markov chains, transmission bottlenecks) for extensions beyond n-grams. `filecite turn0file0`

Slava Katz (1987). *Estimation of probabilities from sparse data for the language model component of a speech recognizer*. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 35(3):400–401. ¹¹²

Summary: Introduces a principled approach to probability estimation under sparse counts, motivating back-off and smoothing ideas that became central in n-gram LMs. Offers historical grounding for why sparse-support models are sensitive to rare events—mirroring drift-driven tail loss in recursive loops. ¹¹³

Methodology: Theoretical/methods paper on n-gram probability estimation under sparse observations. ¹¹³

Key findings: Sparse-data estimation requires carefully designed back-off/smoothing; naive MLE can assign zero probability to unseen events, a property that amplifies tail fragility. ¹¹⁴

Limitations: Predates modern neural LMs; does not address recursive retraining or selection filters. ¹¹⁵

Relevance: Supports the manuscript’s use of unsmoothed/back-off n-grams as a “worst-case transparent” regime for observing drift and support collapse. `filecite turn0file0`

Reinhard Kneser & Hermann Ney (1995). *Improved backing-off for m-gram language modeling*. *ICASSP 1995*:181–184. ¹¹⁶

Summary: Develops improved back-off distributions for n-gram language modeling under sparse data, historically associated with Kneser–Ney smoothing. Demonstrates that the choice of back-off distribution materially affects model quality—relevant to whether drift manifests as hard tail deletion versus softened redistribution. ¹¹⁷

Methodology: Language-modeling methods paper with theoretical derivations and empirical evaluations in speech/NLP contexts. ¹¹⁸

Key findings: Back-off should be optimized for continuation distributions rather than raw frequency; improved smoothing can reduce sparse-data pathologies. ¹¹⁷

Limitations: Does not address recursive training loops; smoothing reduces but does not eliminate distributional shifts induced by selection or by repeated resampling. ¹¹⁹

Relevance: Provides an immediate methodological extension for the manuscript: testing whether smoothing changes the drift-selection trade-off (e.g., less “hard” tail loss but potentially different equilibrium behavior). `filecite turn0file0`

Alison Etheridge (2009). *Some Mathematical Models from Population Genetics* (lecture notes). ¹²⁰

Summary: Presents the Wright–Fisher model and related stochastic processes as canonical models of drift in finite populations. Clarifies assumptions (discrete generations, sampling properties) that map directly onto “finite resampling” interpretations in recursive corpora. ¹²

Methodology: Theoretical lecture notes for stochastic-process models in population genetics. ¹²⁰

Key findings: Finite sampling alone can drive fixation/loss outcomes; adding mutation/selection changes equilibria and stability, providing a mature vocabulary for “drift/selection/innovation” decomposition. ¹²¹

Limitations: Biological setting; mapping requires caution because text generation and learning are not literal reproduction, and selection mechanisms in text can be algorithmically engineered rather than natural. ¹²²

Relevance: Tightens the manuscript’s core analogy and suggests modeling extensions (mutation as fresh human data; selection as verifier/ranker; population size as effective sample size). `filecite{turn0file0}`

Synthesis across themes, methods, and evidence

The literature converges on a shared mechanism-level story, but differs sharply on which components are emphasized and on which regimes are treated as realistic.

The most robust cross-paper consensus is that **iterative self-consumption under replacement** drives contraction: rare forms disappear first (tails, low-support structures), outputs become more “typical,” and diversity metrics fall over iterations. This is explicit in model-collapse demonstrations for text and images and is echoed in linguistic-diversity analyses that measure decline across recursive finetuning. ¹²³ The manuscript’s drift theorem and its Conan Doyle/Austen/Darwin contraction examples are a transparent instantiation of this same qualitative trend, but with exact computability and an explicit decomposition into drift versus selection. `filecite{turn0file0}`

A second consensus is that **selection/verification can stabilize outputs** in the sense of improving correctness or short-horizon viability, but that it can also reduce diversity and suppress traces. Verification-based selection (e.g., selecting synthesized samples that pass tests or constraints) is repeatedly proposed as a mitigation for collapse and is empirically supported in controlled tasks. ¹²⁴ The manuscript’s “lookahead selection” offers a general acceptance-conditioned template for these mechanisms, and its discussion of “soft vs hard publication rules” anticipates the practical importance of graded filters rather than brittle, discontinuous accept/reject boundaries. `filecite{turn0file0}`

Where the literature becomes controversial is in whether “collapse” is **inevitable** in the real internet ecosystem. One line of work implies inevitability under broad conditions, emphasizing information loss and tail deletion under recursion. ¹²⁵ Another line argues that the common “replace” assumption is too pessimistic: if data accumulates rather than replaces, collapse can be avoided, at least in certain modalities and training setups. ³⁴ This disagreement is not merely semantic; it’s a dispute over the correct *state variables* for the ecosystem (fixed-size corpus vs growing archive; proportion of fresh “mutation” vs endogenous synthetic reproduction), and it supports the manuscript’s insistence on “drift and selection” rather than a single binary verdict. `filecite{turn0file0}`

A third, increasingly important line of evidence concerns **real-world selection in public text**. Platform ranking and curation can causally reweight political content exposure and engagement, changing what becomes visible and thus what plausibly enters the durable record. ¹²⁶ In the manuscript’s language, these are ecosystem-scale “selection operators” that modify the public conditional distribution, and they may interact with drift-driven tail loss by systematically downweighting low-engagement or minority content.

¹²⁷

Finally, the “artefact-learning vs process-learning” distinction points to a gap that is not yet well integrated into collapse debates. Inference-time selection methods (self-consistency) and training-time preference

selection (RLHF) often publish only final answers, while intermediate attempts are hidden. ¹²⁸ Meanwhile, process supervision results show that step-level signals can outperform outcome-only supervision for complex reasoning, implying that suppressing traces may harm process-learning even when final-answer quality improves. ³ The manuscript’s theoretical framing directly targets this missing connection: selection reshapes the public conditional, and what is excluded (failed traces) may be exactly what enables future learners to learn search. `filecite{turn0file0}`

A compact publication timeline that reflects these threads is:

```

timeline
  title Key publications shaping drift/selection + recursive training debates
  1951 : Shannon - prediction/entropy of English
  1987 : Katz - sparse data + back-off estimation
  1995 : Kneser & Ney - improved back-off (Kneser-Ney)
  2006 : Cover & Thomas - data-processing inequality in information theory
  2014 : Kirby et al. - iterated learning & language evolution (review)
  2021 : Dodge et al. - documenting C4; filtering effects & bias
  2021 : Bender et al. - stochastic parrots; data governance framing
  2022 : Wei et al. - chain-of-thought prompting
  2022 : Ouyang et al. - RLHF/InstructGPT (selection via preference)
  2023 : Lightman et al. - process supervision; PRM800K
  2023 : Alemohammad et al. - MAD in self-consuming loops (arXiv)
  2023 : Bohacek & Farid - nepotistic training collapse (images)
  2024 : Shumailov et al. - model collapse (Nature)
  2024 : Guo et al. - decline of linguistic diversity (NAACL Findings)
  2024 : Bertrand et al. - stability with clean/synthetic mixtures (ICLR)
  2024 : Gerstgrasser et al. - replace vs accumulate regimes
  2024 : FineWeb; DataComp-LM - dataset curation as selection operators
  2025 : Feng et al. - verification to prevent collapse (ICLR)
  2025 : Nadas et al. - survey of LLM-based synthetic data generation

```

Methodological appraisal and practical recommendations

The dominant methods in this area fall into five families, each with characteristic strengths and blind spots.

Interpretable recursion models (n-grams, Markov chains, Wright–Fisher analogues) isolate mechanisms and make drift/selection mathematically legible, but they risk oversimplifying key real-world features (nonstationary web content, tokenization, retrieval augmentation, long context, and multi-objective training). Yet these models are uniquely good at making “public conditional distributions” explicit, which is difficult with large transformers, and thus they are well-suited for the manuscript’s goal of separating drift from selection. `filecite{turn0file0}`

Deep iterative retraining experiments provide high external validity for modern generative models but often conflate multiple factors: data replacement vs accumulation, filtering/dedup steps, sampling temperature/decoding, and evaluation leakage. ¹²⁹ A recurring limitation is that “collapse” is diagnosed via

broad performance metrics, while distributional diagnostics (tail mass, diversity, representational coverage) are secondary; Guo et al. is a notable exception with purpose-built diversity metrics. ³⁸

Verifier/selection-based mitigation studies are strong when correctness is checkable (math, code, constrained generation), but weaker for open-ended natural language where verifiers may encode subjective preferences or platform incentives. ¹³⁰ The manuscript’s distinction between “soft” and “hard” publication rules is highly relevant here: graded selection can encourage stability without discontinuous mode switching, and it also provides a tunable knob for exploring diversity–viability trade-offs. `filecite turn0file0`

Dataset curation benchmarks and audits show that filtering and deduplication are powerful—and that they can disproportionately remove minority language or topics—implying that “selection” is both performance-relevant and ethically consequential. ¹³¹ This connects directly to the manuscript’s claim that filtering can standardize artefacts while erasing rare forms and minority discourse. `filecite turn0file0`

Computational social science studies of recommender systems give causal evidence that selection mechanisms reshape consumption and exposure at scale, supporting the manuscript’s “public record is filtered” premise. ¹²⁶ They suggest that any long-run theory of public text should treat ranking/engagement feedback as endogenous, not exogenous.

Recommendations for strengthening and extending the manuscript’s literature-grounded contribution follow directly from these observations.

First, explicitly treat **regime design** as a primary experimental variable: at minimum, compare replace vs accumulate loops (as in Gerstgrasser et al.) within the same modeling framework, not only for perplexity/support but also for diversity and tail retention. ¹³² In the manuscript’s terms, “a replacement” is one knob; “archive growth vs fixed-length corpus” is another. `filecite turn0file0`

Second, broaden outcome measures beyond fluency/support to include **multi-level diversity and tail diagnostics**. Guo et al.’s lexical/syntactic/semantic diversity framing is a natural template; Dodge et al.’s dataset documentation lenses (provenance; what gets excluded) can motivate which tails matter socially. ³⁵ Concretely, reporting (a) vocabulary/type retention, (b) n-gram type counts across orders, (c) entropy rate estimates, and (d) distributional divergence (e.g., KL/Jensen–Shannon) between urtext and later generations would triangulate drift. ¹³³

Third, formalize “selection” operationally using a **menu of filters** that map onto real pipelines: best-of-n/self-consistency style selection, verifier-based acceptance, and platform-like engagement proxies. ¹³⁴ The manuscript already provides a general acceptance-conditioning equation; aligning each experimental filter to a real-world analogue strengthens external validity. `filecite turn0file0`

Fourth, connect the manuscript’s “process-learning” hypothesis to empirical evidence by including at least one controlled task where intermediate traces are meaningful and evaluable. Lightman et al.’s process-supervision results and PRM800K offer both conceptual support and a concrete dataset paradigm; contrasting “final-answer-only” versus “trace-rich” corpora in a verifier-rich domain gives a direct test of the manuscript’s claim that filtering redistributes resource-bounded information between artefacts and processes. ¹³⁵

Fifth, leverage modern dataset infrastructures to test selection hypotheses at scale in a reproducible way. DataComp-LM provides a controlled environment for varying filtering/dedup/mixing strategies; FineWeb provides a documented large-scale filtering pipeline whose design choices can be mapped to “selection strength.” ¹³⁶ Pairing these with the manuscript’s interpretable toy models can create a two-level argument: mechanism-level theory + dataset-scale instantiation. □filecite□turn0file0□

Annotated bibliography table designed for sorting

The table below is structured so it can be sorted by Year, Methodology, and Relevance (copy/paste into a spreadsheet for interactive sorting). Bibliographic and methodological details are supported by the annotated entries above.

Year	Short citation	Primary domain	Work type	Methodology (tags)	Central construct(s)	Relevance to manuscript (1–5)
1951	Shannon, <i>Prediction and entropy of printed English</i>	Info theory / LM origins	Foundational	Human prediction + theory	Context, entropy, prediction	2
1987	Katz, <i>Sparse data estimation for LM</i>	Speech/ NLP	Foundational	Estimation / back-off	Sparse counts, back-off	3
1995	Kneser & Ney, <i>Improved backing-off</i>	NLP	Foundational	Smoothing theory + evals	Back-off distributions	3
2006	Cover & Thomas, <i>Elements of Information Theory</i>	Info theory	Foundational	Theory	Data-processing inequality	3
2009	Etheridge, <i>Models from population genetics</i>	Population genetics	Foundational	Stochastic processes	Wright–Fisher drift/selection	4
2014	Kirby et al., <i>Iterated learning & language evolution</i>	CogSci	Review	Review (multi-method)	Iterated learning chains	3

Year	Short citation	Primary domain	Work type	Methodology (tags)	Central construct(s)	Relevance to manuscript (1–5)
2021	Dodge et al., <i>Documenting C4</i>	NLP / datasets	Empirical	Dataset audit	Filtering effects, bias, contamination	5
2021	Bender et al., <i>Stochastic parrots</i>	NLP ethics	Conceptual	Critical synthesis	Data governance, harms of scale	3
2022	Wei et al., <i>Chain-of-thought prompting</i>	LLM reasoning	Empirical	Prompting experiments	Visible traces, reasoning steps	4
2022	Ouyang et al., <i>RLHF/ InstructGPT</i>	Alignment	Empirical	SFT + reward modeling + RLHF	Preference-based selection	4
2022	Huszár et al., <i>Algorithmic amplification on Twitter</i>	Comp social sci	Empirical	Large RCT + causal inference	Ranking as selection	4
2023	Wang et al., <i>Self-consistency</i>	LLM reasoning	Empirical	Sampling + selection	Trace selection before output	4
2023	Lightman et al., <i>Verify step by step</i>	LLM reasoning	Empirical	Process supervision + verifier	Step traces, PRM800K	5
2023	Bohacek & Farid, <i>Nepotistic training collapse</i>	GenAI (vision)	Empirical	Iterative retraining (images)	Cross-domain collapse evidence	3
2024	Shumailov et al., <i>Model collapse</i>	GenAI	Empirical + theory	Theory + multi-model demos	Tail loss, collapse	5
2024	Alemohammad et al., <i>MAD</i>	GenAI (vision)	Empirical + theory	Loop taxonomy + experiments	Quality-diversity trade-off	4

Year	Short citation	Primary domain	Work type	Methodology (tags)	Central construct(s)	Relevance to manuscript (1–5)
2024	Bertrand et al., <i>Stability of iterative retraining</i>	GenAI (vision)	Theory + empirical	Stability conditions + experiments	Clean fraction for stability	4
2024	Gerstgrasser et al., <i>Replace vs accumulate</i>	GenAI (multi-modal)	Empirical + theory	Iterative pretraining + analysis	Accumulation avoids collapse	5
2024	Guo et al., <i>Linguistic diversity decline</i>	NLP	Empirical	Recursive finetuning + diversity metrics	Diversity erosion	5
2024	Penedo et al., <i>FineWeb</i>	NLP datasets	Empirical (datasets)	Web-scale curation + ablations	Filtering as selection	4
2024	Li et al., <i>DataComp-LM</i>	NLP datasets	Benchmark	Controlled dataset experiments	Filtering and mixing	4
2025	Feng et al., <i>Verification prevents collapse</i>	GenAI	Theory + empirical	Verifier theory + tasks	Selection via verification	5
2025	Nadas et al., <i>Synthetic data generation survey</i>	NLP	Review	Survey	Synthetic pipelines + verification	3

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